KILNDECK · DESIGN DECISION RECORD

KilnDeck decisions of 2026-10-30

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DOCUMENT CONTROL

Document ID	KND-DDR-004	Version	0.1
Status	Draft	Date	2026-10-30
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/kilndeck		
Source	docs/decisions/0004-decisions-2026-10-30.md @ 9f92bd8		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-30	Amish Chadha	Records D10 (concept A), D11 (wind working limit) and D14 (33 kg team-handling limit) as decided by Amish on 2026-10-30, what was carried through, and the shortfall found (end truss modules 34.4 and 34.8 kg, proposed as D16)

0004: Decisions of 2026-10-30

- **Date:** 2026-10-30
- **Status:** accepted (D10, D11, D14); D16 proposed, awaiting Amish

Context

After a report listing the proposed calls in four repos, Amish wrote on 2026-10-30: "on the calls - i go with your recommendations". The three open decisions of KND-DEC-001 each carried a stated recommendation, so each is decided as recommended.

What was decided

#	Decision	Option taken	Carried through
D10	Concept carried forward	A, the rolling bridge deck (152 of 190, 18 % ahead of the wall-top cableway at 124)	`concept_chosen: A`; KND-DDR-003 accepted; no design change
D11	Wind working limit	(a) plus (b): work up to 7 m/s, clamp and tie down above it and before leaving the deck; and clamp and tie down at every shift end whatever the weather	Build plan stop 8 and step 18; PR-004 and the assumptions in KND-PRS-001; KND-CAL-001 s15; `sizing.py` prints the working limit
D14	Team-handling limit for pieces carried up the kiln	(b) the HSE two-person guide, about 33 kg: five truss modules a side (1,380 mm) and 2.4 m track modules	`cad/src/model.py` (`MODULES` 4 to 5, `TRACK_LEN` 3,000 to 2,400, `TRACK_N` 4 to 5, `PAD_PITCH` 1,000 to 800 so every joint keeps a pad); PR-012 and PR-016; BOM; drawings; build plan

What changed in the numbers (D14)

Item	Before	After
Truss modules	8 of 1,725 mm	10 of 1,380 mm
Track modules	8 of 3,000 mm	10 of 2,400 mm
Pads	26	32
Floor panels, bearers, pillow blocks, carriage rails	8, 9, 3, 4	10, 11, 4, 5
Heaviest piece	39.8 kg (end truss module)	34.8 kg (end truss module)
Deck mass	810 kg	853 kg
Pad pressure, working load and rated load	0.151 and 0.207 MPa	0.154 and 0.209 MPa (interim limit 0.20)
BOM total	USD 2,786.97	USD 2,904.67 (about USD 118 more, all estimates scaled from earlier prices)

`budget_usd` stays at 3,500.

What the recommendation did not foresee

The recommendation said every piece would be under 33 kg. The detailed model gives: track modules 26.7 kg, middle truss modules 32.9 kg, floor panels 24.8 kg, outer end truck 29.8 kg, but the end truss modules 34.4 kg (side A) and 34.8 kg (side B), because each carries a 10 mm foot plate of 1.9 kg. PR-012 is therefore recorded as not met by 1.4 to 1.8 kg, and STEM rows S17 and S22 as fails with a fix proposed. This does not change the working principle, pitch or scope. Options are proposed in KND-DEC-001 D16: bolt the foot plates on (recommended, no margin), six modules a side, or accept the shortfall.

Not decided

- D16, above: new, no decision.
- The approach to a co-design partner (KND-DDR-001, D8) is an action for Amish, not a proposal with a recommendation.

Consequences

- Option A's model changed, so the options picture (`media/options.png`), the product renders and the cards need re-rendering by the coordinator; they are not made here.
- The wind rule is a rule for the operator; the independent review found the hand force used up at about 5 m/s on a 2 % grade, so a graded kiln needs a lower limit, which the partner survey sets.